

120 Years of Olympic History: Participation and Medal Trends

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Introduction

Olympic Games:

- ▶ The most popular international sports competition every four years, with over 200 nations participate, with over 10,000 athletes, and over 300 events
- ▶ Have expanded enormously in scale, diversity and global importance

Studying historical trends in Olympic participation and performance can shed light on wider trends in gender equality, national development and sporting investment. Analysed a data-based view on the evolution of the Olympic movement.

Dataset + Research Questions

Data of athletes, events, medals, and attributes (Summer and Winter Olympics), collected from Kaggle, from 1896 to 2016 (120 years of Olympic history).

- ▶ How has overall athlete participation in the Summer Olympic Games changed over time?
- ▶ How has female representation among Olympic athletes changed over time?
- ▶ Has the proportion of Olympic medals won by female athletes increased over time?
- ▶ Has the distribution of Olympic medals among nations become more or less concentrated over time?

Methodology

Visualisation of trends in participation and medals over time

Implementation of a simple linear regression model, logistic regression model and HHI to test specific hypotheses about the relationship

1. Overall participation patterns over time:

▶ $Participation_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$

2. Female Medal Success over time:

▶ $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 * Year + \epsilon_t$

3. Medal Concentration Among Nations:

▶ $HHI_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$

Results

Visualisation

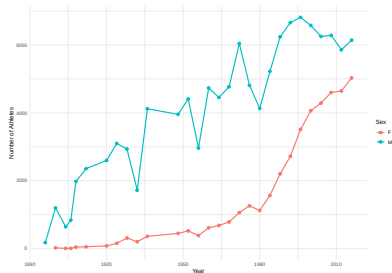


Figure 1: Athlete Participation by Gender in the Summer Olympics (1896–2016)

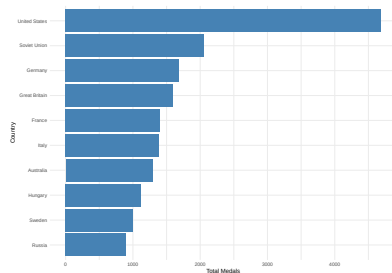


Figure 2: Top 10 Countries by Total Medal Count in the Summer Olympics (1896–2016)

Visualisation

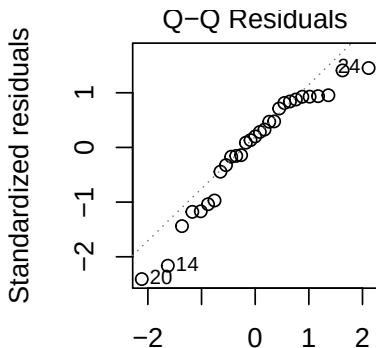
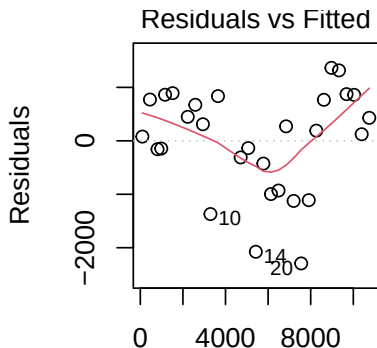
Table 1: Top 5 Countries by Total Medals (Summer Olympics)

Country	Total Medals
United States	4686
Soviet Union	2061
Germany	1687
Great Britain	1598
France	1408

Overall participation patterns over time

Table 2

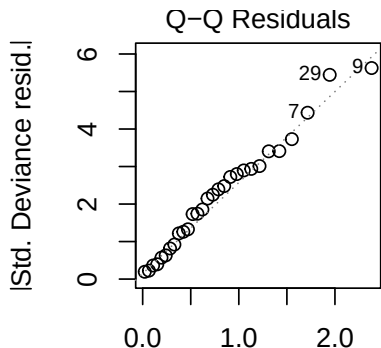
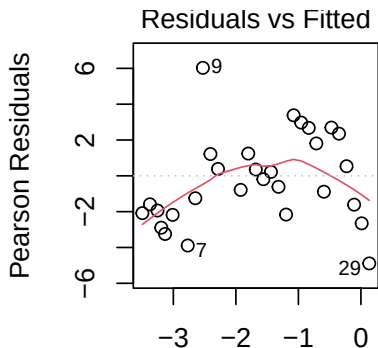
Term	Coefficient	p-value	R ²
Intercept	96.6205	0.7825	0.924784415470121
Year	88.7696	0.0000	



Female Medal Success over time

Table 3

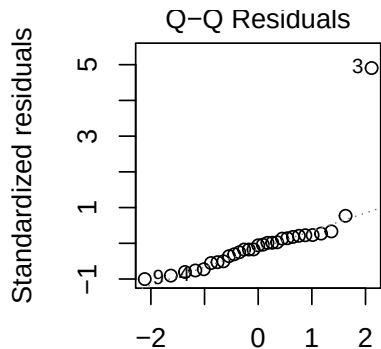
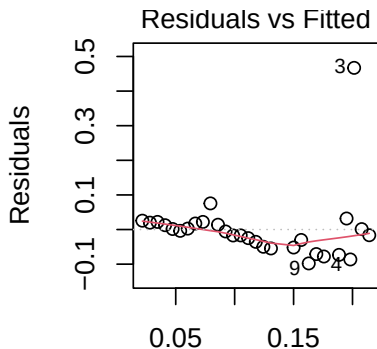
Term	Coefficient	p-value	Pseudo R ²
Intercept	-3.4954	0	0.964539393668614
Year	0.0302	0	



Medal Concentration Among Nations

Table 4

Term	Coefficient	p-value	R ²
Intercept	0.2142	0.0000	0.274710131339156
Year	-0.0016	0.0035	



Conclusion and Recommendation

- ▶ Female representation among Olympic athletes has increased (visualisation), and the proportion of medals won by female athletes has also increased significantly over time
- ▶ The distribution of Olympic medals across nations has become less concentrated over time

Limitation: Not account for other factors and the linear and logistic regression models not fully capture non-linear changes

Future research: : Explore more complex models and conduct qualitative analyses to understand the social and political factors driving changes in Olympic participation and success